

NCAA March Madness Prediction System

Project Update #2: Model Development, Barttorvik Integration & Backtest Results

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1 Overview

We spent most of Weeks 2–4 building models and watching them lose to logistic regression. Nine models, three tiers of complexity, leave-one-tournament-out backtests on 11 seasons. The turning point came when we scraped Barttorvik’s continuous efficiency ratings [Barttorvik, 2017] and fed them into the same logistic model that had been stuck at 0.196 Brier for two weeks. It dropped to 0.159. That’s below Vegas (~ 0.175) and below our proposal target of < 0.170 , and we got there not by building a better model but by finding a better feature.

2 Models

2.1 Baselines

We started with the obvious ones. Seed logistic ($P = \sigma(\beta_0 + \beta_1 \Delta s)$) gives 0.199. KenPom logistic, same form on POM ordinal rank difference, gives 0.196. These two anchored our expectations for the first two weeks: anything below 0.196 counts as progress.

We also built an Elo rating system ($K = 24$, 75-point home-court, 50% inter-season regression, margin-of-victory multiplier). It scored 0.221. We tried tuning the parameters but the problem is more basic than that. Elo only knows who won each game and by how much. It cannot distinguish a 10-point win over Houston from a 10-point win over a team ranked 250th. The MOV adjustment helps at the margins but Elo’s information set is just too narrow.

We also tried a logistic model on our own self-computed tempo-adjusted efficiency (50 iterations of SOS fixed-point adjustment from box scores). 0.225. Worse than POM ordinal rank, which was discouraging. Our SOS model averages box scores and iterates; KenPom and Barttorvik use regression frameworks with home-court effects, tempo, and player availability data. The gap shows.

The model that changed everything was Barttorvik logistic: a single feature, the continuous net rating difference ($\text{AdjOE} - \text{AdjDE}$), in a logistic regression. 0.169 Brier. One continuous number outperformed every ordinal-rank model we had, including XGBoost on 53 features. We probably should have tried this two weeks earlier.

2.2 Advanced models

XGBoost was our biggest disappointment before Barttorvik, and our biggest improvement after. The initial version scored 0.214 on 53 features, worse than a one-variable logistic. We had 27 rating columns from 9 Massey systems, all correlated above 0.95 with each other, and the model was splitting on this noise with 670 training games per fold. After adding 5 Barttorvik features, dropping 29 redundant columns, and tightening regularization (depth $4 \rightarrow 3$, trees $300 \rightarrow 150$, L1/L2 penalties), it came down to 0.172.

The Multi-Feature Logistic takes 21 features (seed, POM rank diff, all 5 Barttorvik diffs, self-computed efficiency diffs, 8 Four Factors diffs, momentum, coach experience) through a

StandardScaler into logistic regression. 0.159 Brier. We tried various values of C and settled on $C = 0.5$. The relationship between team quality difference and win probability is close enough to sigmoidal that a linear model in log-odds space does fine once you give it the right inputs.

The Mixture of Experts routes matchups to 4 specialized logistic experts based on seed gap (Chalk, Competitive, Toss-up, catch-all), trained via EM. 0.185, improved from 0.209 after Barttorvik features and a scaling fix from Week 2. The experts do learn different behaviors, but the gating doesn't add much over a single model.

We also rebuilt the Transformer from scratch. The Week 2 version (4 layers, 64-dim, no pretraining) scored 0.245, which was embarrassing. The new version uses 2 layers, 32-dim, about 8K parameters. We pretrain the encoder on ~5K regular-season game pairs per season, freeze it, train the matchup MLP on tournament data, then unfreeze with a $10\times$ smaller learning rate. 0.196 Brier. Pretraining made the difference between a model that learns nothing and one that matches the KenPom baseline, but it still can't touch the Barttorvik models because it doesn't have access to cross-team adjusted efficiency in its game embeddings.

2.3 Ensemble

We took 4 models (Multi-Feature Logistic, Barttorvik Logistic, Seed Logistic, XGBoost), trained them per LOTO fold, and ran an inner leave-one-season-out loop to optimize blend weights via `scipy`. 0.161 Brier. Not much help. The optimizer puts 80–95% weight on Multi-Feature Logistic in every fold, meaning the ensemble is basically the logistic model with a tiny correction. We ran it anyway because we wanted to confirm there wasn't hidden complementary signal in the other models. There isn't.

3 Barttorvik Integration

This deserves its own section because it was the turning point. We built a scraping pipeline that fetches T-Rank data for all D1 teams for each season from 2014 to 2025. The site uses JavaScript browser verification, so the scraper posts a `js_test_submitted` form field to get through. Team names from Barttorvik don't match Kaggle's TeamIDs exactly ("Miami FL" vs "Miami (FL)", "NC State" vs "North Carolina St", that sort of thing), so we fuzzy-match against the `MTeamSpellings.csv` mapping using `rapidfuzz` for anything that doesn't match directly. Results get cached to `data/external/barttorvik_{season}.csv`.

Five matchup features come out: net rating diff ($\text{AdjOE} - \text{AdjDE}$), offensive efficiency diff, defensive efficiency diff, Barthag (Pythagorean win probability) diff, and tempo diff. The reason these work so much better than ordinal ranks is simple. A team with AdjOE of 118.5 and one at 114.2 differ by 4.3 points per 100 possessions. That 4.3 is information. Ordinal ranks throw it away: both teams might be ranked in the top 20, five spots apart, but the rank difference of 5 is the same whether the efficiency gap is 0.3 or 8.0.

Going from POM ordinal (0.196) to Barttorvik net rating (0.169) was a 0.027-point improvement from changing one feature in the same logistic model. Adding the remaining Barttorvik features plus Four Factors, momentum, and coach data into a 21-feature logistic pushed it to 0.159.

4 Backtest Results

Table 1 has the full numbers.

Everything above the line at 0.185 uses Barttorvik. Everything below it doesn't. 2025 was the best season for every single model because the bracket played out mostly to seed (three of the Final Four were 1 or 2 seeds). 2021 was the hardest for most models. That was the first tournament after COVID canceled 2020 entirely, so rosters were unusual, non-conference

Table 1: LOTO Brier scores (2014–2025, 11 folds). Coin flip = 0.250, seed baseline $\approx$ 0.200, Vegas $\approx$ 0.175.

Model	Mean Brier	Std	Best / Worst
Multi-Feature Logistic	0.159	0.020	2025 / 2023
Weighted Ensemble	0.161	0.021	2025 / 2014
Barttorvik Logistic	0.169	0.022	2025 / 2018
XGBoost (regularized)	0.172	0.025	2025 / 2014
Mixture of Experts	0.185	0.023	2025 / 2023
KenPom Logistic	0.196	0.017	2025 / 2021
Transformer (pretrained)	0.196	0.024	2025 / 2014
Seed Logistic	0.199	0.021	2025 / 2021
Elo (MOV-adjusted)	0.221	0.029	2014 / 2021
Efficiency Logistic	0.225	0.016	2015 / 2022

schedules were limited, and the selection committee was working with less information than normal.

The year-to-year noise is worth paying attention to. Standard deviations run around 0.02. That means our “best model” at 0.159 will look like a 0.180 model in a wild tournament and a 0.140 model in a chalky one. A single 16-over-1 upset (UMBC over Virginia in 2018) costs about 0.01 Brier for the entire season. In a 67-game tournament, one game is roughly 1.5% of your score.

5 Discussion

The main thing we took away from these three weeks is that feature quality matters more than we expected and model complexity matters less. We built an MoE, a Transformer, an ensemble with inner-loop weight optimization. The thing that actually worked was plugging a better number into logistic regression.

In hindsight this shouldn’t have been surprising. [Pomeroy \[2004\]](#) and past Kaggle winners have said the same thing: the competition is won on features, not models. But we had to learn it ourselves. We spent two weeks stuck at 0.196, trying fancier architectures, before realizing the ceiling was in the data, not the model.

A few other observations from the backtest. XGBoost with 53 features (0.214) loses to logistic regression on 1 feature (0.196). That’s not because XGBoost is a bad algorithm. It’s because 27 of those 53 features are near-duplicates, and with only 670 training games per fold, the model fits the redundancy instead of the signal. After we cut to 24 curated features it dropped to 0.172. Still not as good as logistic on the same features (0.159), but respectable. Tree models overfit more easily on small datasets, and 670 tournament games is very small.

The Transformer’s trajectory is also worth noting. It went from 0.245 without pretraining to 0.196 with pretraining on regular-season data. That’s the difference between a model that can’t learn from 736 examples and one that can, because the encoder comes into fine-tuning already knowing what a “strong team” looks like from having seen 50,000 regular-season games. But 0.196 still only matches the KenPom logistic baseline, because the Transformer’s game embeddings don’t include the cross-team efficiency adjustments that Barttorvik provides. It knows a team’s FG% and turnover rate in each game, but it doesn’t know what those numbers mean relative to the quality of the opponent’s defense. We suspect adding Barttorvik features to the game embeddings would close this gap, and that’s on the list for the final phase.

The ensemble at 0.161 barely moves the needle over the best single model at 0.159. The optimizer consistently puts almost all weight on Multi-Feature Logistic. That probably means we’ve extracted most of what these features have to offer. Getting below 0.150 would likely

require player-level data (who's injured, who's in foul trouble historically, matchup-specific defensive stats) or proprietary information we don't have access to.

6 Next Steps

The final phase runs through May 1. We want to try injecting Barttorvik features into the Transformer's game embeddings to see if temporal modeling and efficiency data can combine for something better than either alone. We'll also wire up the Bayesian market-calibration module (already implemented but unused) with historical betting lines. And we need to generate actual 2026 bracket predictions for the Kaggle submission. Our 0.159 is competitive with past top-10 Kaggle entries, so we're cautiously optimistic, but a lot depends on whether 2026 is a chalk year or a chaos year.

References

- Oliver, D. *Basketball on Paper: Rules and Tools for Performance Analysis*. Potomac Books, 2004.
- Pomeroy, K. KenPom.com: College Basketball Ratings. <https://kenpom.com>, 2004–present.
- Barttorvik. T-Rank: College Basketball Ratings. <https://barttorvik.com>, 2017–present.
- Sokol, J., Jank, W., and Harp, S. LRMC: Logistic Regression / Markov Chain for NCAA Basketball. Georgia Institute of Technology, 2006–2026.